Hints: Euclidean geometry – 1. Vectors, angles, simplices and volumes in $\mathbb{R}^n$

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Theorem 3 by induction on n, as in the sketch. Project onto $v_m^\perp$: $\langle w_i, w_j\rangle = \langle v_i, v_j\rangle - \frac{\langle v_i, v_m\rangle\langle v_j, v_m\rangle}{|v_m|^2} \leqslant \langle v_i, v_j\rangle$, because both $\langle v_i, v_m\rangle, \langle v_j, v_m\rangle \leqslant 0$. If two of the w_i vanished, two of the v_i would be negative multiples of v_m and then positive multiples of each other, i.e. at an acute angle. So at least $m-2$ non-zero non-acute vectors live in $\mathbb{R}^{n-1}$ and $m-2 \leqslant 2(n-1)$. In the obtuse case no w_i vanishes ($v_i = -cv_m$ would make $\langle v_i, v_j\rangle$ and $\langle v_m, v_j\rangle$ of opposite signs) and $m-1 \leqslant n$.

Hint 2. Examples: $\pm e_1, \dots, \pm e_n$ give $2n$; the $n+1$ vectors from the centre of a regular simplex to its vertices give $n+1$ (their pairwise products are $-\frac{1}{n}$ times R^2, see the homework).

Hint 3. For $2n$ vectors the count is tight at every step, so for each v_m exactly one w_i vanishes: some $v_k = -cv_m$, $c > 0$. For every other v_i we get $\langle v_i, v_m\rangle \leqslant 0$ and $\langle v_i, v_k\rangle = -c\langle v_i, v_m\rangle \leqslant 0$, so $\langle v_i, v_m\rangle = 0$. Thus the remaining $2n-2$ vectors lie in $v_m^\perp$; induct.

Hint 4. For the last claim: $0 \leqslant \left|\sum v_i\right|^2 = m + \sum_{i\neq j}\langle v_i, v_j\rangle \leqslant m - m(m-1)\beta$, so $m \leqslant 1 + \frac{1}{\beta}$.

Problem 2

Hint 1. (a) Put A_0 at the origin and the edges along the coordinate axes. The facet opposite A_0 projects onto the coordinate hyperplane $x_i = 0$ exactly as the facet opposite A_i (the vertex A_i goes to the origin, the others stay). Apply Theorem 2 with $k = n-1$.

Hint 2. (b) Fix a unit vector u and project S orthogonally onto $u^\perp$. By Theorem 2 the facet opposite A_i projects to a set of volume $F_i|\langle u_i, u\rangle|$. The facets with $\langle u_i, u\rangle > 0$ cover the projection of S once (they form the "upper" boundary), and so do the facets with $\langle u_i, u\rangle < 0$. Hence $\sum_i F_i\langle u_i, u\rangle = 0$ for every u, i.e. $\sum F_i u_i = 0$.

Hint 3. So $F_0 u_0 = -\sum_{i\geqslant 1} F_i u_i$. Taking lengths gives $F_0 \leqslant \sum_{i\geqslant 1} F_i$, with equality only if all u_i are parallel, which is impossible. Squaring gives $F_0^2 = \sum F_i^2 + 2\sum_{i<j} F_i F_j\langle u_i, u_j\rangle$, and the angle between outer normals of two facets is $\pi - \varphi_{ij}$, so $\langle u_i, u_j\rangle = -\cos\varphi_{ij}$.

Problem 3

Hint 1. Put $v_i = A_i - B$. Since B is interior, $\sum \lambda_i v_i = 0$ with all $\lambda_i > 0$ (barycentric coordinates). Consider the graph on $\{1, \dots, n+1\}$ in which i and j are joined when $\langle v_i, v_j\rangle < 0$, i.e. when $\angle A_i B A_j > 90°$. It suffices to show that this graph is connected.

Hint 2. If not, split the indices into two non-empty parts I, J with $\langle v_i, v_j\rangle \geqslant 0$ for $i \in I$, $j \in J$. Then $w = \sum_{i\in I}\lambda_i v_i = -\sum_{j\in J}\lambda_j v_j$ and $|w|^2 = -\sum_{i\in I, j\in J}\lambda_i\lambda_j\langle v_i, v_j\rangle \leqslant 0$ (the lemma of Theorem 3).

Hint 3. So $\sum_{i\in I}\lambda_i v_i = 0$: B is a convex combination of the vertices A_i, $i \in I$, i.e. it lies on a proper face of the simplex, not inside. A connected graph on $n+1$ vertices has at least n edges.

Problem 4

Hint 1. Work in barycentric coordinates: $M = \sum \lambda_j A_j$ with $\lambda_j > 0$, $\sum \lambda_j = 1$. The point N_i is on the line $A_i M$ and has i-th coordinate 0, so $N_i = \frac{M - \lambda_i A_i}{1-\lambda_i}$, with coordinates $\frac{\lambda_j}{1-\lambda_i}$ ($j \neq i$) and 0 at i.

Hint 2. The ratio of volumes is the absolute value of the determinant of the matrix of barycentric coordinates of $N_1, \dots, N_{n+1}$. This matrix is $D^{-1}(\mathbf{1}\lambda^T - \Lambda)$ with $D = \operatorname{diag}(1-\lambda_i)$, $\Lambda = \operatorname{diag}(\lambda_i)$, and $\det(\mathbf{1}\lambda^T - \Lambda) = \det(-\Lambda)(1 - \lambda^T\Lambda^{-1}\mathbf{1})$. So the ratio equals

$$\frac{n\,\lambda_1 \cdots \lambda_{n+1}}{(1-\lambda_1)\cdots(1-\lambda_{n+1})}.$$

Hint 3. By AM–GM, $1-\lambda_i = \sum_{j\neq i}\lambda_j \geqslant n\big(\prod_{j\neq i}\lambda_j\big)^{1/n}$; multiplying over i gives $\prod(1-\lambda_i) \geqslant n^{n+1}\prod\lambda_j$. Hence the ratio is at most $\frac{1}{n^n}$, with equality exactly for $\lambda_i = \frac{1}{n+1}$: M is the centroid.

Problem 5

Hint 1. Work with signed volumes; since P is inside, all $v(S_i)$ have the sign of $v(S)$. The circumcentre C of a simplex with vertices $X_0, \dots, X_n$ solves the linear system $2\langle C, X_i\rangle + c = |X_i|^2$ $(i = 0, \dots, n)$ in the unknowns $C \in \mathbb{R}^n$, $c \in \mathbb{R}$.

Hint 2. By Cramer's rule, $v(S)\,C(S)$ is, up to a constant factor depending only on n, the vector of determinants D_k of the $(n+1)\times(n+1)$ matrix with rows $(1, X_i, |X_i|^2)$ with the k-th coordinate column deleted, and $v(S)$ itself is (up to a factor) the determinant D with rows $(1, X_i)$. So it suffices to prove that each such determinant satisfies $\sum_i D(S_i) = D(S)$.

Hint 3. Write $P = \sum\lambda_i X_i$ with $\sum\lambda_i = 1$. Then the row of P equals $\sum\lambda_j\,\mathrm{row}(X_j) + (0, \dots, 0, \delta)$ with $\delta = |P|^2 - \sum\lambda_j|X_j|^2$. Replacing row i by it gives $\lambda_i D + \delta\,\mathrm{cof}_i$, where cof_i is the cofactor of the last entry of row i.

Hint 4. Summing over i: $\sum\lambda_i D = D$, and $\sum_i \mathrm{cof}_i$ is the determinant of the matrix whose last column is replaced by all ones; it has two equal columns (the first column is also all ones), so it vanishes.